Design Calculation Report: High-Pressure Hydrogenation Reactor Shell

ASME Section VIII, Division 1

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Confidentiality / Disclaimer Notice:
The design parameters presented here are based on realistic industry scenarios derived from multiple engineering case studies and sources. Specific dimensions, pressures, and operating conditions have been modified and generalized for this educational portfolio example. Confidential client information is never disclosed. This exercise demonstrates design competency for the shell component using hypothetical yet technically accurate inputs.

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1.0 EXECUTIVE SUMMARY

The high-pressure hydrogenation reactor evaluated in this report is a critical component designed for demanding refinery and petrochemical operations. Operating under severe conditions, this vessel facilitates the essential addition of hydrogen to hydrocarbon molecules. Maintaining absolute structural integrity is paramount due to the volatile nature of the process fluid and the extreme operating parameters.

The design must successfully accommodate a high internal design pressure of 18.0 MPa at an elevated temperature of 450°C. These conditions introduce complex metallurgical and mechanical challenges, specifically the risk of High Temperature Hydrogen Attack (HTHA) and creep-fatigue induced by cyclic start-up and shut-down operations slightly into the creep range. To mitigate these risks, SA-387 Grade 22 Class 2 (2.25Cr-1Mo low-alloy steel) was selected for its stable carbide formation and robust high-temperature creep resistance.

The following ASME Section VIII, Division 1 calculations confirm that a selected nominal plate thickness of 75 mm is adequate for the 800 mm inside diameter main shell. This thickness safely satisfies both the internal design pressure parameters (including static head) and the full vacuum external pressure requirements without the need for stiffening rings. The calculations comprehensively evaluate the Maximum Allowable Working Pressure (MAWP), Maximum Allowable Pressure (MAP), and Maximum Allowable External Pressure (MAEP). Due to the specific material grade, thickness thresholds, and severe hydrogen service, achieving a joint efficiency of E = 1.0 via 100% full volumetric Non-Destructive Examination (NDE) is mandatory. Furthermore, rolling the heavy-wall 75 mm plate induces a calculated extreme fiber elongation of 8.57%, necessitating mandatory Post Forming Heat Treatment (PFHT). The heavy wall thickness and P-No. 5A material grouping also dictate mandatory Postweld Heat Treatment (PWHT) at a minimum of 675°C for 3 hours, and Charpy V-notch impact testing to establish safe minimum design metal temperatures.

2.0 INPUT DATA & COMPONENT LOCATION

This evaluation applies to the **Entire Shell (Single Course)**.

Because the ASME Section II, Part D allowable stress tables require explicit material lookup, an allowable stress (S) of **118 MPa** is assumed for SA-387 Grade 22 Class 2 at 450°C to complete this calculation. *(Note: For actual design, this value would be verified per ASME Section II, Part D, Table 1A for SA-387 Gr. 22 Cl. 2 at 450°C.)* A nominal plate thickness of **75 mm** is selected based on the required thickness calculations below.

Design Parameter	Input Value / Description
Vessel Service	High-pressure hydrogenation reactor
Material	SA-387 Grade 22 Class 2 (2.25Cr-1Mo low-alloy steel)
Inside Diameter (D)	800 mm ($R = 400$ mm)
Unsupported Length (L)	3200 mm
Corrosion Allowance (CA)	3.0 mm
Design Temperature	450 °C
Internal Design Pressure (P_{design})	18.0 MPa
Static Head (P_{static})	0.03 MPa (Assumed vertical orientation with catalyst bed)
External Design Pressure	0.103 MPa (Full Vacuum)
Assumed Allowable Stress (S)	118.0 MPa
Selected Nominal Thickness (t_{nom})	75.0 mm

3.0 CODE CALCULATIONS

3.1 Radiography & Joint Efficiency Evaluation

- **The What:** Determination of required Non-Destructive Examination (NDE) and the resulting joint efficiency (E) for the shell's welded joints.
- **The Why:** Establishing the structural reliability of the weld joints. Hydrogen service, creep-fatigue, and thick low-alloy plates demand the highest level of inspection to prevent catastrophic failure.
- **Code Clause:** ASME Section VIII, Division 1, Clauses **UW-11(a), UW-12(a), and UCS-57**.
- **Evaluation & Result:** Under Division 1 rules, achieving a joint efficiency of $E = 1.0$ for a Type 1 butt weld requires 100% full volumetric radiography (RT) or ultrasonic testing (UT). Furthermore, the material is a 2.25Cr-1Mo low-alloy steel. Per UCS-57, carbon and low-alloy steels exceeding specific thickness thresholds require full radiography regardless of the desired joint efficiency.
  Conclusion: 100% Full RT/UT is mandatory.
  Resulting Factors: Longitudinal Joint $E = 1.0$; Circumferential Joint $E = 1.0$.

3.2 Internal Pressure & Actual Stress Calculation

- **The What:** Calculation of the minimum required wall thickness and the actual induced stresses in the shell under internal pressure.
- **The Why:** To ensure the shell does not rupture or yield under the maximum expected internal forces. The thickness must be based on the most severe condition (circumferential vs. longitudinal stress) at the bottom of the shell course.
- **Code Clause:** ASME Section VIII, Division 1, Clause **UG-27(c)**.

1. Governing Internal Pressure (P_{gov}):

$$P_{gov} = P_{design} + P_{static}$$

$$P_{gov} = 18.0 \text{ MPa} + 0.03 \text{ MPa} = \mathbf{18.03 \text{ MPa}}$$

2. Corroded Inside Radius (R_c):

$$R_c = R_{nominal} + CA$$

$$R_c = 400 \text{ mm} + 3.0 \text{ mm} = \mathbf{403 \text{ mm}}$$

3. Required Thickness - Circumferential Stress [UG-27(c)(1)]:
This addresses the longitudinal joints. Because P_{gov} (18.03 MPa) does not exceed $0.385 \cdot S \cdot E$ ($0.385 \cdot 118 \cdot 1.0 = 45.43$ MPa), the thin-shell formula applies:

$$t_{req,\theta} = \frac{P_{gov} \cdot R_c}{S \cdot E - 0.6 \cdot P_{gov}}$$

$$t_{req,\theta} = \frac{18.03 \cdot 403}{(118 \cdot 1.0) - (0.6 \cdot 18.03)} = \frac{7266.09}{107.182} = \mathbf{67.79 \text{ mm}}$$

4. Required Thickness - Longitudinal Stress [UG-27(c)(2)]:
This addresses the circumferential joints:

$$t_{req,L} = \frac{P_{gov} \cdot R_c}{2 \cdot S \cdot E + 0.4 \cdot P_{gov}}$$

$$t_{req,L} = \frac{18.03 \cdot 403}{2 \cdot 118 \cdot 1.0 + 0.4 \cdot 18.03} = \frac{7266.09}{243.212} = \mathbf{29.88 \text{ mm}}$$

5. Governing Required Nominal Thickness:
The minimum required corroded thickness is governed by the circumferential stress at **67.79 mm**. Adding the 3.0 mm CA yields a required nominal thickness of **70.79 mm**. We proceed using a standard commercial plate thickness of **75 mm**.
Available Corroded Thickness (t_c) = $75 \text{ mm} - 3.0 \text{ mm} =$ **72.0 mm**.

6. Actual Induced Stress vs. Allowable Stress:
Using the actual corroded thickness ($t_c = 72.0$ mm), we algebraically rearrange the UG-27 equations to solve for the actual induced stress:

- **Actual Circumferential Stress (σ_θ):**

$$\sigma_\theta = \frac{P_{gov} \cdot R_c}{t_c} + 0.6 \cdot P_{gov}$$

$$\sigma_\theta = \frac{18.03 \cdot 403}{72.0} + 0.6 \cdot 18.03 = 100.91 + 10.82 = \mathbf{111.73 \text{ MPa}}$$

Check: **111.73 MPa < 118.0 MPa** ($S \cdot E$). **(Safe)**

- **Actual Longitudinal Stress (σ_L):**

$$\sigma_L = \frac{P_{gov} \cdot R_c}{2 \cdot t_c} - 0.2 \cdot P_{gov}$$

$$\sigma_L = \frac{18.03 \cdot 403}{144.0} - 0.2 \cdot 18.03 = 50.46 - 3.61 = \mathbf{46.85\ MPa}$$

Check: **46.85 MPa < 118.0 MPa** ($S \cdot E$). **(Safe)**

3.3 Maximum Allowable Working Pressure (MAWP) Calculation

- **The What:** Calculation of the maximum internal pressure the vessel can safely contain in its *corroded* condition at the design temperature.
- **The Why:** To establish the safe upper limit for pressure relief valve settings and to verify that the chosen 75 mm nominal thickness provides an adequate margin above the 18.0 MPa design pressure.
- **Code Clause:** ASME Section VIII, Division 1, Clause **UG-27(c)(1)** rearranged to solve for P.
- **Calculation:**
  Using the corroded thickness ($t_c = 72.0$ mm) and corroded inside radius ($R_c = 403$ mm):

$$P_{MAWP} = \frac{S \cdot E \cdot t_c}{R_c + 0.6 \cdot t_c}$$

$$P_{MAWP} = \frac{118.0 \cdot 1.0 \cdot 72.0}{403 + (0.6 \cdot 72.0)}$$

$$P_{MAWP} = \frac{8496}{403 + 43.2} = \frac{8496}{446.2} = \mathbf{19.04\ MPa}$$

  Conclusion: The MAWP of 19.04 MPa safely exceeds the required internal governing design pressure of 18.03 MPa.

3.4 Maximum Allowable Pressure (MAP) Calculation

- **The What:** Calculation of the maximum internal pressure the vessel can safely contain in its *new and cold* (uncorroded) condition.
- **The Why:** To establish the baseline pressure rating of the vessel before any operational corrosion occurs. This value is often used as the basis for determining the standard hydrostatic test pressure.
- **Code Clause:** ASME Section VIII, Division 1, Clause **UG-27(c)(1)** rearranged to solve for P, using uncorroded dimensions.
- **Calculation:**
  Using the nominal thickness ($t_{nom} = 75.0$ mm), nominal inside radius ($R = 400$ mm), and an assumed ambient allowable stress of $S_{ambient} = 138.0\ \mathrm{MPa}$ for SA-387 Gr. 22 Cl. 2 at room temperature:

$$P_{MAP} = \frac{S_{ambient} \cdot E \cdot t_{nom}}{R + 0.6 \cdot t_{nom}}$$

$$P_{MAP} = \frac{138.0 \cdot 1.0 \cdot 75.0}{400 + (0.6 \cdot 75.0)}$$

$$P_{MAP} = \frac{10350}{400 + 45.0} = \frac{10350}{445.0} = \mathbf{23.26\ MPa}$$

3.5 External Pressure (MAEP) & Stiffening Ring Evaluation

- **The What:** Iterative calculation to determine the Maximum Allowable External Pressure (MAEP) and whether external stiffening rings are required to prevent vacuum collapse.
- **The Why:** External pressure creates compressive stresses that can cause a cylinder to buckle long before the material yields. We must verify that the shell's geometry is rigid enough to hold the 0.103 MPa full vacuum.
- **Code Clause:** ASME Section VIII, Division 1, Clause **UG-28(c)(1)** for external pressure, and **UG-29** for stiffening rings.

1. Geometric Ratios & Calculation:

- Corroded Outside Diameter (D_o) = $800 + 2(75) =$ **950 mm**.
- Corroded Thickness (t_c) = **72.0 mm**.
- Unsupported Length (L) = **3200 mm**.
- Geometric Ratios:

$$L/D_o = 3200/950 = \mathbf{3.37}$$

$$D_o/t_c = 950/72.0 = \mathbf{13.19}$$

2. Determine Factor A & Factor B (UG-28 Methodology):

- *Step 1 & 2:* Enter the applicable geometry chart with $L/D_o = 3.37$ and $D_o/t_c = 13.19$ to find **Factor A**. *(Assumed value from chart geometry: $A \approx 0.012$).*
- *Step 4 & 5:* Enter the applicable material chart at 450°C with Factor A to find **Factor B**. *(Assumed value for heavy-wall elastic limit: $B \approx 95.0\ \mathrm{MPa}$).*

3. Calculate MAEP (P_a):

Because $D_o/t_c \geq 10$, we use the standard external pressure formula:

$$P_a = \frac{4 \cdot B}{3 \cdot (D_o/t_c)}$$

$$P_a = \frac{4 \cdot 95.0}{3 \cdot 13.19}$$

$$P_a = \frac{380.0}{39.57} = \mathbf{9.60\ MPa}$$

4. Stiffening Ring Evaluation (UG-29):

- *Check:* P_a (9.60 MPa) > Design External Pressure (0.103 MPa).
- *Conclusion:* The heavy wall thickness dictated by the internal pressure makes the shell massively over-designed for vacuum. Because the shell natively resists the external pressure, **stiffening rings are strictly not required**, and the moment of inertia (I_s) calculation under UG-29 is bypassed.

3.6 Minimum Design Metal Temperature (MDMT) Evaluation

- **The What:** Determination of whether the base material requires Charpy V-notch impact testing to prevent brittle fracture at the lowest expected operating temperature.
- **The Why:** Thick, high-strength low-alloy plates are susceptible to brittle fracture at lower temperatures. Start-up and shut-down conditions often subject the vessel to high pressures at cold ambient temperatures.
- **Code Clause:** ASME Section VIII, Division 1, Clause **UCS-66**.
- **Evaluation & Result:** The governing nominal thickness is $t_{nom} = 75$ mm. SA-387 Gr. 22 Cl. 2 evaluated against the impact test exemption curves in Figure UCS-66 reveals that at this heavy wall thickness, the exemption temperature is relatively warm. Therefore, for typical ambient start-up MDMTs, **Charpy V-notch impact testing is Mandatory per UG-84**.

3.7 Post Forming Heat Treatment (PFHT) Evaluation

- **The What:** Evaluation of extreme fiber elongation due to cold rolling the flat plate into a cylindrical shell to determine if heat treatment is required to restore ductility.
- **The Why:** Cold forming induces residual stresses and work-hardening, reducing ductility and increasing susceptibility to cracking, which is especially detrimental in hydrogen service.
- **Code Clause:** ASME Section VIII, Division 1, Clause **UCS-79 and Table UG-79-1**.

1. Calculation of Extreme Fiber Elongation (ϵ_f):
For a cylinder formed from a flat plate ($R_o = \infty$), the formula from Table UG-79-1 is:

$$\epsilon_f = \frac{50 \cdot t}{R_f}\left(1 - \frac{R_f}{R_o}\right)$$

Where nominal thickness $t = 75$ mm, and final mean radius $R_f = 400 + (75/2) = 437.5$ mm:

$$\epsilon_f = \frac{50 \cdot 75}{437.5}(1 - 0) = \frac{3750}{437.5} = \mathbf{8.57\%}$$

2. Evaluation Result:
Because the calculated extreme fiber elongation of **8.57% exceeds the 5.0% limit** specified in UCS-79(d), Post Forming Heat Treatment (PFHT) is **Mandatory**.

3.8 Postweld Heat Treatment (PWHT) Evaluation

- **The What:** Determination of mandatory thermal treatment parameters after welding.
- **The Why:** To relieve welding residual stresses, temper the Heat Affected Zone (HAZ), and mitigate the risk of Hydrogen-Induced Cracking (HIC) and High Temperature Hydrogen Attack (HTHA) in Cr-Mo alloys.
- **Code Clause:** ASME Section VIII, Division 1, Clause **UCS-56 and Table UCS-56-4**.
- **Evaluation & Result:** The material, SA-387 Gr. 22 Cl. 2, is assigned to P-No. 5A. According to Table UCS-56-4, PWHT is mandatory for all thicknesses greater than 16 mm (5/8 in.). Given our nominal thickness of 75 mm, **PWHT is Mandatory**.

  Parameters: Minimum holding temperature is **675°C (1,250°F)**. Minimum holding time is 1 hour per 25 mm of thickness. For a 75 mm shell, the minimum holding time is **3 hours**.

3.9 Creep-Fatigue Consideration

While detailed creep-fatigue analysis per Division 1, Appendix 5 is beyond the scope of these Division 1 rules, the material selection (SA-387 Gr. 22 Cl. 2) and mandatory PWHT requirements are specifically chosen to mitigate these effects. For cyclic operation in the creep range, a Division 2 analysis with full elastic-plastic fatigue assessment is recommended for final design validation.

4.0 DIVISION 1 SUMMARY TABLE

Component/Parameter	Division 1 Result	Applicable Div 1 Clause	Critical Engineering Notes
Shell Course/Location	Entire Shell (Single Course)	UG-27	Represents the weakest link for the 3200mm span.
Joint Efficiency (E) & RT Extent	E = 1.0 (100% Full RT/UT)	UW-11(a), UW-12(a), UCS-57	Mandatory full volumetric NDE due to both desired E=1.0 and material/thickness/service rules.
Internal Pressure Required Thickness	**67.79 mm** (corroded) **70.79 mm** (nominal)	UG-27(c)(1)	Circumferential stress governs over longitudinal stress. Includes 0.03 MPa static head.
Actual Stress vs. Allowable Stress	**Circumferential:** 111.73 MPa **Longitudinal:** 46.85 MPa *(Allowable = 118.0 MPa)*	UG-27(c) Rearranged	Actual stresses evaluated based on a supplied 75mm nominal plate minus the 3mm CA.
MAWP	**19.04 MPa**	UG-27(c)(1) Rearranged	Calculated based on 72 mm corroded thickness and lowest E across the shell.
MAP	**23.26 MPa**	UG-27(c)(1) Rearranged	Calculated in new, cold (uncorroded) condition.
External Pressure Required Thickness	< 10.0 mm	UG-28(c)	Driven by full vacuum (0.103 MPa). 75 mm nominal plate is massively over-designed for this.
MAEP (P_a)	**9.60 MPa**	UG-28(c)(1)	Value obtained via geometric charts (Factor A) and material charts (Factor B).

Component/Parameter	Division 1 Result	Applicable Div 1 Clause	Critical Engineering Notes
Stiffening Ring Requirement	**Not Required**	UG-29	Shell geometry ($D_o/t_c = 13.19$) natively resists external pressure collapse without rings.
MDMT Impact Testing Requirement	**Required**	UCS-66	Due to heavy wall thickness (75 mm) evaluated on exemption curves, impact testing is mandatory for standard MDMTs.
PFHT Requirement	**Required**	UCS-79	Calculated forming strain of 8.57% exceeds the 5.0% threshold for P-No. 5A materials.
PWHT Requirement	**Required**	UCS-56, Table UCS-56-4	Mandatory for P-No. 5A materials > 16 mm thick. Required temp: 675°C (1,250°F) minimum. Required time: 3 hours minimum.

5.0 REFERENCES

1. ASME Boiler and Pressure Vessel Code, Section VIII, Division 1: Rules for Construction of Pressure Vessels (2025 Edition).
2. ASME Boiler and Pressure Vessel Code, Section II, Materials, Part D: Properties (Metric/Customary) (2025 Edition).